HYDROPURE

Transforming water

STANDARD OPERATING PROCEDURE

FOR

OPERATION AND MAINTENANCE OF PW GENERATION PLANT

CLIENT NAME	GINNI FILAMENTS LTD.
LOCATION	PANOLI
SYSTEM	PURIFIED WATER GENERATION SYSTEM
DOCUMENT NO.	HSPL/GFL/PR681/SOP/PWG/21

MANUFACTURE AND SUPPLIER

Hydropure System Pvt. Ltd.

412, Diamond Industrial Estate,

Dahisar East, Mumbai 400 068, INDIA

Tele Fax: +91 22 289 78725, 2897 9097

www.hydropure.in

R-0	**27-08-2021**	**FIRST DRAFT**	
REV NO.	**ISSUE DATE**	**DESCRIPTION**	**REMARKS**

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1.0 PURPOSE

To define the procedure for operation and maintenance of the PW generation plant used to generate purified water

2.0 SCOPE

This SOP is applicable to process generation plant. Details of system are as below

MAKE	CAPACITY	EQUIPMENT CODE	LOCATION
Hydropure Systems Pvt. Ltd.	3500 LPH	___	Generation Water Room

3.0 REFERENCES, ATTACHMENTS AND ANNEXURE

3.1 REFERENCE

3.1.1 RO Manual

3.1.2 Control Panel Operation

3.1.3 Control Philosophy for PW Generation System

3.1.4 P & ID Diagram

3.2 ATTACHMENT

3.2.1 Attachment–1 Equipment Log Sheets

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3.2.7 Attachment–7 EDI CIP Records

3.2.8 Attachment–8 Status Label

3.3 ANNEXURE

3.3.1 **ANNEXURE – 1** SOP of Chemical Solution Preparation

This document elaborated chemical solution preparation procedure and has calculation sheet which states data about inline flow, original chemicals to be used, preparation of desired solution to be dosed, details of dosing pump and setting of flow rates of dosing pumps

This document also elaborates frequency of preparation of chemical solution

Attachment – I of this document gives reference that how operator should generate **'Monthly Log Sheet - Preparation of Chemical Solution'** to enable him to keep records for this activity (**This document is attached as 'Attachment 01 to this SOP)**

3.3.2 **ANNEXURE – 2** Maintenance Schedule sheet

This document elaborates maintenance schedule of major equipments and some of the critical instruments, which need regular precautionary measures apart from calibration

4.0 RESPONSIBILITIES

4.1 OPERATOR

4.1.1 To operate the PW Generation plant as per define procedure

4.1.2 To maintain records in log sheets

4.2 SUPERVISOR

4.2.1 To ensure that operation of PW Generation plant is carried out as per define procedure

4.2.2 To provide training to operator

4.2.3 To ensure the cleaning, operation & preventive maintenance of PW Generation plant is carried out as per define procedure

4.2.4 To ensure that proper records are maintain

4.3 DEPARTMENT HEAD

4.3.1 To ensure that proper operation & maintenance is carried as per SOP

4.3.2 To ensure implementation of SOP

4.4 QUALITY ASSURANCE DEPARTMENT

4.4.1 To ensure that the job is carried out as per procedure

4.4.2 To ensure that proper records are maintain

4.5 QUALITY HEAD

4.5.1 To review and approve new or revised SOP's

4.6 FACTORY HEAD

4.6.1 To review and approve new or revised SOP's

5.0 DISTRIBUTION

5.1 Engineering Department

5.2 Quality Assurance Department

6.0 DEFINITION OF TERMS

6.1	ADV	:	Actuated Diaphragm Valve
6.2	AV	:	Auto Valve
6.3	Bar	:	Bar is a unit of pressure equal to one million (10^6) dynes, equivalent to 10 newtons, per square centimeter
6.4	BFV	:	Butterfly Valve
6.5	BV	:	Ball Valve
6.6	CF	:	Cartridge Filter
6.7	CIP	:	Clean In Place
6.8	CIPFP	:	CIP pump
6.9	CIPT	:	CIP Tank
6.10	CPG	:	Compound Pressure Gauge
6.11	DP	:	Dosing Pump
6.12	DT	:	Dosing Tank
6.13	DV	:	Diaphragm Valve
6.14	EVF	:	Electrical Vent Filter
6.15	FLS	:	Float Type Level Switch
6.16	FS	:	Flow Switch
6.17	HMI	:	Human Machine Interface
6.18	HSPL	:	Hydropure System Pvt. Ltd.
6.19	LLS	:	Low Level Switch
6.20	LPH	:	Litres Per Hour
6.21	LS	:	Level Switch
6.22	M^3/hr	:	Unit of volume, Cubic Meter Per Hour
6.23	MSDS	:	Material Safety Data Sheet
6.24	MWC	:	One of pressure Unit. Meters of Water Column(10.197 MWC is 1 bar pressure)
6.25	NRV	:	Non Return Valve
6.26	GFL	:	GINNI FILAMENTS LTD.
6.27	PG	:	Pressure Gauge
			PH = Log_{10} 1/(H^+) = -Log_{10} (H^+)’

6.28	pH	:	PH is defined as the logarithm of reciprocal of the hydrogen ion concentration or is equal to the logarithm of the hydrogen ion concentration with negative sign
6.29	PID	:	Piping and Instrumentation Diagram
6.30	PLC	:	Programmable Logic Controller
6.31	PQ	:	Performance Qualification
6.32	PS	:	Pressure Switch
6.33	PSIG	:	PSIG is a unit of pressure as pounds per square inch gauge
6.34	PT	:	Pressure Transmitter
6.35	PTS	:	Pretreatment System
6.36	PU	:	Polyurethane
6.37	QA	:	Quality Assurance
6.38	RECP	:	Recirculation Pump
6.39	RM	:	Rotameter
6.40	ROFP	:	RO Feed Pump
6.41	SDI	:	Silt Density Index
6.42	SM	:	Static Mixer
6.43	UB	:	‘U’ Shaped Bend
6.44	UPVC	:	Unplasticized polyvinyl chloride

7.0 PROCEDURE

7.1 SAFETY PRECAUTIONS

7.1.1 Read 'Operating and Maintenance Manual' of individual equipment carefully before proceeding with operation of pretreatment system. Safety precautions specified in these manuals shall be followed carefully. Do not deviate from the data specified in these manuals

7.1.2 While using any cleaning chemical, MSDS of respective chemical shall be handy before using it. Read its handling and safety procedures carefully before handling them

7.1.3 Ensure that all cleaning chemicals are dissolved and well mixed before circulating the cleaning solution through the system

7.1.4 Refer 'Dosing Solution Preparation' SOP while preparing online dosing chemical solutions (ANNEXURE – 1)

7.1.5 Wear hand gloves and face mask while handling chemicals

7.1.6 Follow safety processes published by respective EHS department

7.2 POSITION OF VALVES

Following procedures are to be followed during working of the system

7.2.1 POSITION OF MANUAL VALVES – Position of manual valves recommended in below table will remain same for 'Auto' as well as 'Manual' mode of operations

Refer below table for the position of manual valves of PW Generation system-

Position of manual valves specified in below table refers to service mode (either in AUTO or MANUAL mode) i.e. operation of PW Generation System for the production of purified water to suffice downstream requirement

Position of some of these valves may change during CIP (Clean-In-Place) of individual equipment. The same will be specified in respective section

DO NOT FORGET TO CHANGE position of respective downstream valves of equipment during cleaning operation. OTHERWISE it may cross contaminate downstream system

SR. NO.	MANUAL VALVE TAG NO.	DESCRIPTION OF VALVE	POSITION OF MANUAL VALVE (DURING SERVICE MODE)
A	**SERVICE MODE OF RO FEED SYSTEM**		
1.	DV – 201	At bottom of RO Feed Tank	Close
2.	DV – 202	Suction valve of RO Feed Pump ROFP – 201	Open
3.	DV – 203	Discharge valve of RO Feed Pump ROFP – 201	Open Partially*
4.	DV-204	Discharge Recirculation valve of RO Feed Pump ROFP – 201	Open Partially*
5.	NV – 201	Isolation valve for pH & ORP Sensors	Open Partially*
B	**SERVICE MODE OF RO-I SYSTEM**		

SR. NO.	MANUAL VALVE TAG NO.	DESCRIPTION OF VALVE	POSITION OF MANUAL VALVE (DURING SERVICE MODE)
6.	DV – 205	Discharge valve of RO High Pressure Pump ROHP – 201	Open Partially*
7.	NV – 202	RO reject to recirculation at Feed line of RO Pass-I	Open Partially*
C	**SERVICE MODE OF RO-II SYSTEM**		
8.	DV – 206	Discharge valve of RO High Pressure Pump ROHP – 202	Open Partially*
9.	NV-203	Reject to recirculation at Feed line of RO Pass-II	Open Partially*
D	**SERVICE MODE EDI SYSTEM**		
10.	DV – 207	Concentrate feed of EDI System EDI–201	Open Partially*
E	**SAMPLING VALVES OF COMPLETE SYSTEM**		
11.	Sampling valves shall be closed throughout the operation (Service mode or cleaning) of system. Sampling valves shall only be opened for collection sampling		

*** This setting will be done by Supplier's personnel during commissioning and shall not be changed unless and until there is change in feed water quality or downstream demand. Contact supplier before making any changes**

Discharge valve of any pump shall be opened partially before starting system if this setting needs to be changed. Then start the system (Auto Mode or Manual Mode) and adjust this valve to get the desired recommended flow to avoid any damage to downstream equipment

7.2.2 POSITION OF AUTO VALVES –

7.2.2.1 **In 'Auto Mode' -** operation of auto valves will be taken care by PLC control panel

7.2.2.2 **In 'Manual Mode' -** auto valves of PW Generation system shall be operated manually through HMI screens. Sequence of this operation shall be listed in 'Operation of system in Manual Mode' section of this document

7.3 CHEMICAL SOLUTIONS FOR ONLINE DOSING

7.3.1 REFER –

7.3.1.1 ANNEXURE – 1 , 'SOP of Chemical Solution Preparation' to prepare chemical solution

7.3.1.2 ATTACHMENT – 1 for 'Chemical Solution Preparation log sheet'

7.3.2 Refer ANNEXURE – 1 of SOP of Chemical Solution Preparation' to verify settings of dosing pumps, concentration of chemical solution to be prepared

7.3.3 Online dosing chemicals shall be prepared daily

7.3.4 Discard last day's left over chemical and clean dosing tanks properly before adding freshly prepared solution

7.3.5 Chemical solutions for CIP (Clean-In-Place) shall be freshly prepared while performing CIP of RO system. Left over solution shall be discarded immediately after CIP. Clean the tank and keep it ready for next use

8.0 HOT WATER SANITIZATION OF RO-I, RO-II & EDI SYSTEM

8.1 CRITERIA FOR HOT WATER SANITIZATION:

8.1.1 If any parameter out of microbial OR TOC, goes beyond maximum allowable limit specified by regulatory guidelines

8.1.2 **OR** as a schedule maintenance activity as specified in **point no. 8.2.1**

8.2 HOT WATER SANITIZATION FREQUENCY:

8.2.1 Once in a **THREE MONTHS ± 1 week**

8.3 HOT WATER SANITIZATION TIME:

8.3.1 30-45 Minutes (Can be validated in PQ phase)

8.4 PREREQUISITES

8.4.1 Ensure that all required utilities are connected and available at respective terminal

8.4.2 Display warning sign board stating ‘HOT SURFACE DO NOT TOUCH’ near RO system.

E.g. 

8.4.3 Inform end users about sanitization of system so that they will schedule their work accordingly

8.4.4 STOP RO-I, RO-II & EDI System if it is working in AUTO / MANUAL mode

8.4.5 Fill the RO Feed tank (T-201) with water **<u>up to HIGH level (‘H’ of LS-201)</u>** for sanitization of RO-I, RO-II & EDI system.

8.5 SANITIZATION OF RO-I, RO-II & EDI SYSTEM PROCEDURE

NOTE: System will be in 'Idle / standby Mode' before putting in 'Sanitization Mode', hence all auto valves will be in **OPEN OR CLOSED** position as per the control philosophy

8.5.1 Put the system in sanitization mode by touching the icon SANITIZATION MODE

8.5.2 After selection of SANITIZATION MODE icon below window will pop-up for selecting the system.

8.5.3 Now system will initiate sanitization of RO-I, RO-II & EDI System in auto mode as mentioned in Control Philosophy document of PW generation system

8.5.4 System will generate an alarm with message 'RO Sanitization Cycle Over' & after completion of sanitization

8.5.5 System abort sanitization with message and alarm in some conditions as specified in control philosophy of PW generation system

8.5.6 Refer 'Control Philosophy' of 'PW Generation System' for details of hot water sanitization of system

8.5.7 Control system will drain water during after sanitization till 'LOW LOW' level of LS – 201

8.5.8 Hence drain the tank (T-201) manually after sanitization, by opening the drain valve (DV-201) completely

8.5.9 Now RO-I, RO-II & EDI System is ready to put in service mode.

9.0 CHEMICAL CLEANING (CIP) OF RO & MEMBRANES

- CIP shall be avoided as far as possible hence maintain the good health during service mode
- RO permeate shall be flushed for an hour after completion of CIP
- Then RO permeate may be used for production purpose.
- But RO needs stabilization of 6 – 12 hours after CIP, hence result of CIP shall be concluded after 12 hours of continuous operation

9.1 CRITERIA FOR CIP:

9.1.1 Permeate flow dropped by 10-15% at the same feed pressure

9.1.2 Feed pressure increases by 10 -15% for same permeate flow

9.1.3 10-15% increase in pressure drop across RO system

9.1.4 OR After 6 month ± 7 days

9.2 CIP FREQUENCY:

9.2.1 Below mentioned duration OR above mentioned criteria in point no. 9.1 whichever is earlier

9.2.2 For Chemically Sanitizable RO System (CSRO), 3 months ± 1 week

9.2.3 For Hot Water Sanitizable RO System (HSRO), 6 months ± 1 week

9.3 CHEMICALS USED:

9.3.1 For scheduled CIP of RO membranes as per the frequency mentioned in point no. 9.2.2 perform the CIP with the Citric acid at pH 4.5

9.3.2 For CIP with any criteria found as mentioned in point no. 9.1, perform the CIP with the chemical solutions as mentioned in point no. 9.7.5 & 97.6

9.4 PREREQUISITES

9.4.1 Ensure that all required utilities are connected and available at respective terminal

9.4.2 Inform end users about CIP of system so that they will schedule their work accordingly

9.4.3 Display warning sign board stating 'CIP IN PROGRESS DO NOT TOUCH' near RO System.

E.g. 

9.4.4 STOP PW Generation system if it is working in AUTO / MANUAL mode

9.4.5 Make CIP system, RO Feed Tank (T-201), RO Feed Pump (ROHP-201) ready for CIP of RO system.

9.4.6 Fill the RO Feed Tank (T-201) with water **up to HIGH level** for CIP of RO & EDI system.

9.5 SAFETY PRECAUTIONS

Safety precautions shall be followed while doing chemical cleaning of RO membranes. In addition to maintaining adequate ventilation, personal protective equipment shall be utilized to prevent injury or personnel hazards.

9.6 CLEANING REQUIREMENT

In normal operation, the membrane in reverse osmosis elements can become fouled by mineral scale, biological matter, colloidal particles and insoluble organic constituents. Deposits build up on the membrane surfaces during operation until they cause loss in normalized permeate flow, loss of normalized salt rejection, or both.

Elements should be cleaned when one or more of the above mentioned parameters are applicable. If we wait too long, cleaning may not restore the membrane element performance successfully. In addition, the time between cleanings becomes shorter as the membrane elements will foul or scale more rapidly.

Differential Pressure (ΔP) should be measured and recorded across each stage of the array of pressure vessels. If the feed channels within the element become plugged, the ΔP will increase. It should be noted that the permeate flux will drop if feed water temperature decreases. This is normal and does not indicate membrane fouling.

A malfunction in the pretreatment, pressure control, or increase in recovery can result in reduced product water output or an increase in salt passage. If a problem is observed, these causes should be considered first. The element(s) may not require cleaning

9.7 PROCEDURE

This SOP provides general information about the usual foulants affecting the performance of Reverse Osmosis (RO) membrane elements and the removal of these foulants.

Note: The Composite Polyamide type of RO membrane elements may not be exposed to chlorinated water under any circumstances. Any such exposure will cause irreparable damage to the membrane. Absolute care must be taken following any disinfection of piping or equipment or the preparation of cleaning or storage solutions to ensure that no trace of chlorine is present in the feed water to the RO membrane elements. If there is any doubt about the presence of chlorine, perform chemical testing to make sure. Neutralize any chlorine residual with a sodium bisulfite solution, and ensure adequate mixing and contact time to accomplish complete de-chlorination. Dosing rate is 1.8 to 3.0 ppm sodium bisulfite per 1.0 ppm of free chlorine.

Note: The use of cationic surfactant should be avoided in cleaning solutions, since irreversible fouling of the membrane elements may occur.

9.7.1 RO MEMBRANE FOULING AND CLEANING

During normal operation over a period of time, RO membrane elements are subject to fouling by suspended or sparingly soluble materials that may be present in the feed water. Common examples of foulants are:

- Calcium carbonate scale
- Sulfate scale of calcium, barium or strontium
- Metal oxides (iron, manganese, copper, nickel, aluminium, etc.)
- Polymerized silica scale
- Inorganic colloidal deposits
- Mixed inorganic/organic colloidal deposits
- NOM organic material (Natural Organic Matter)

- Man-made organic material (e.g. antiscalant/dispersants, cationic polyelectrolytes)
- Biological (bacterial bioslime, algae, mold, or fungi)

The nature and rapidity of fouling depends on a number of factors, such as the quality of the feed water and the system recovery rate. Typically, fouling is progressive, and if not controlled early, will impair the RO membrane element performance in a relatively short time. Cleaning is should occur when the RO shows evidence of fouling, just prior to a long-term shutdown, or as a matter of scheduled routine maintenance. The elements shall be maintained in a clean or "nearly clean" condition to prevent excessive fouling by the foulants listed above.

Cleaning should be carried out before these values are exceeded to maintain the elements in a clean or "nearly clean" condition. Effective cleaning is evidenced by the return of the normalized parameters to their initial, Start-up, value. In the event you do not normalize your operating data, the above values still apply if you do not have major changes in critical operating parameters. The operating parameters that have to stay constant permeate flow, permeate back-pressure, recovery, temperature, and feed TDS. If these operating parameters fluctuate, then it is highly recommended that you normalize the data to determine if fouling is occurring or if the RO is actually operating normally based on the change in critical operating parameters.

Monitoring overall plant performance on a regular basis is an essential step in recognizing when membrane elements are becoming fouled. Performance is affected progressively and in varying degrees, depending on the nature of the foulants. Below table "RO Troubleshooting Matrix" provides a summary of the expected effects that common foulants have on performance.

RO cleaning frequency due to fouling will vary by site. If you have to clean more than once a month, you should be able to justify further capital expenditures for improved RO pre-treatment or a re-design of the RO operation. If the cleaning frequency is every one to three months, you may want to focus on improving the operation of your existing equipment but further capital expenditure may be harder to justify.

It is important to clean the membranes when they are only lightly fouled, not heavily fouled. Heavy fouling can impair the effectiveness of the cleaning chemical by impeding the penetration of the chemical deep into the foulants and in the flushing of the foulants out of the elements. If normalized membrane performance drops 30 to 50%, it may be impossible to fully restore the performance back to baseline conditions.

When inorganic or polyelectrolyte coagulants are used in the pre-treatment process, there can often be incomplete reaction of the coagulant and thus insufficient formation of a filterable floc. The user should ensure that excessive amounts of coagulant are not fed to the RO system, as it can lead to fouling. Polyelectrolyte fouling can often be very difficult to remove and result in higher than expected feed pressure. Excessive amounts of inorganic coagulant can be measured by using SDI filter equipment. In the case of iron, the iron on the SDI filter pad should typically be 3 µg/pad and never above 5 µg/pad. In regards to polymer coagulants, the user should discuss the concern with their chemical supplier and have them ensure that the chemical will not adversely affect the membrane.

In addition to the use of turbidity and SDI, particle counters are also very effective to accurately measure the suitability of the feed water for NF/RO elements. The measure of particles greater than 2 microns in size should be < 100 particles per millilitre.

One RO design feature that is commonly over-looked in reducing RO cleaning frequency is the use of RO permeates water for flushing foulants from the system. Soaking the RO elements during standby with permeate can help dissolve scale and loosen precipitates, reducing the frequency of chemical cleaning.

What you clean for can vary site by site depending on the foulants. Complicating the situation frequently is that more than one foulants can be present, which explains why cleanings frequently require a low pH and high pH cleaning regiment.

IMPORANT:
The membrane elements shall not be exposed to feed water containing oil, grease, or other foreign matter which proves to chemically or physically damage the integrity of the membrane.

RO TROUBLESHOOTING
(Pressure Drop is defined as the Feed pressure minus the Concentrate pressure)

POSSIBLE CAUSE	POSSIBLE LOCATION	PRESSURE DROP	FEED PRESSURE	SALT PASSAGE
Metal Oxide Fouling (e.g. Fe,Mn,Cu,Ni,Zn)	1st stage lead elements	Rapid increase	Rapid increase	Rapid increase
Colloidal Fouling (organic and/or	1st stage lead elements	Gradual increase	Gradual increase	Slight increase

POSSIBLE CAUSE	POSSIBLE LOCATION	PRESSURE DROP	FEED PRESSURE	SALT PASSAGE
inorganic complexes)				
Mineral Scaling (e.g. Ca, Mg, Ba, Sr)	Last stage tail elements	Moderate Increase	Slight increase	Marked increase
Polymerized Silica	Last stage tail elements	Normal to increased	Increased	Normal to increased
Biological Fouling	Any stage, usually lead elements	Marked increase	Marked increase	Normal to increased
Organic Fouling (dissolved NOM)	All stages	Gradual increase	Increased	Decreased
Antiscalant Fouling	2nd stage most severe	Normal to increased	Increased	Normal to increased
Oxidant damage (e.g Cl2, ozone,KMnO4)	1st stage most severe	Normal to decreased	Decreased	Increased
Hydrolysis damage (out of range pH)	All stages	Normal to decreased	Decreased	Increased
Abrasion damage (carbon fines, etc)	1st stage most severe	Normal to decreased	Decreased	Increased
O-ring leaks (at interconnectors or adapters)	Random (typically at feed adapter)	Normal to decreased	Normal to decreased	Increased
Glue line leaks (due to permeate back-pressure in service or standby)	1st stage most severe	Normal to decreased	Normal to decreased	Increased
Glue line leaks (due to closed permeate valve while cleaning or flushing)	Tail element of a stage	Increased (based on prior fouling & high delta P)	Increased (based on prior fouling & and high delta P)	Increased

9.7.2 DISCUSSION ON FOULANTS:

Calcium Carbonate Scale: Calcium carbonate is a mineral scale and may be deposited from almost any feed water if there is a failure in the antiscalant/dispersant addition system or in the acid injection pH control system that results in a high feed water pH. An early detection of the resulting calcium carbonate scaling is absolutely essential to prevent the damage that crystals can cause on the active membrane layers. Calcium carbonate scale that has been detected early can be removed by lowering the feed water pH to between 3.0 and 5.0 for one or two hours. Longer resident accumulations of calcium carbonate scale can be removed by a low pH cleaning with a citric acid solution.

Calcium, Barium & Strontium Sulphate Scale: Sulphate scale is a much "harder" mineral scale than calcium carbonate and is harder to remove. Sulphate scale may be deposited if there is a failure in the antiscalant/dispersant feed system or if there is an over feed of sulphuric acid in pH adjustment. Early detection of the resulting sulphate scaling is absolutely essential to prevent the damage that crystals can cause on the active membrane layers. Barium and strontium sulphate scales are particularly difficult to remove as they are insoluble in almost all cleaning solutions, so special care should be taken to prevent their formation.

Calcium Phosphate Scale: This scale is particularly common in municipal waste waters and polluted water supplies which may contain high levels of phosphate. This scale can generally be removed with acidic pH cleaners. At this time, phosphate scaling calculations are not performed by the RO Design software's. As a rule of thumb, contact membrane manufacturers technical department if phosphate levels in the feed are 5 ppm or higher.

Metal Oxide/Hydroxide Foulants: Typical metal oxide and metal hydroxide foulants are iron, zinc, manganese, copper, aluminium, etc. They can be the result of corrosion products from unlined pipes and tanks, or result from the oxidation of the soluble metal ion with air, chlorine, ozone, potassium permanganate, or they can be the result of a pre-treatment filter system upset that utilizes iron or aluminium-based coagulant aids.

Polymerized Silica Coating: A silica gel coating resulting from the super-saturation and polymerization of soluble silica can be very difficult to remove. It should be noted that this type of silica fouling is different from silica-based colloidal foulants, which may be associated with either metal hydroxides or organic matter. Silica scale can be very difficult to remove by traditional chemical cleaning methods. Contact technical department if the traditional methods are unsuccessful. There does exist harsher cleaning chemicals, like ammonium biflouride, that have been used successfully at some sites but are considered rather hazardous to handle and can damage equipment.

Colloidal Foulants: Colloids are inorganic or mixed inorganic/organic based particles that are suspended in water and will not settle out due to gravity. Colloidal matter typically contains one or more of the following major components: iron, aluminium, silica, sulphur, or organic matter.

Dissolved NOM Organic Foulants: The sources of dissolved NOM (Natural Organic Matter) foulants are typically derived from the decomposition of vegetative material into surface waters or shallow wells. The chemistry of organic foulants is very complex, with the major organic components being either humic acid or fulvic acid. Dissolved NOMs can quickly foul RO membranes by being absorbed onto the membrane surface. Once absorption has occurred, then a slower fouling process of gel or cake formation starts. It should be noted that the mechanism of fouling with dissolved NOM should not be confused with the mechanism of fouling created by NOM organic material that is bound up with colloidal particles.

Microbiological Deposits: Organic-based deposits resulting from bacterial slimes, fungi, molds, etc. can be difficult to remove, particularly if the feed path is plugged. Plugging of the feed path makes it difficult to introduce and distribute the cleaning solutions. To inhibit additional growth, it is important to clean and sanitize not only the RO system, but also the pre-treatment, piping, dead-legs, etc. The membranes, once chemically cleaned, will require the use of approved biocide and an extended exposure requirement to be effective. For further information on biocides, refer to Technical Service Technical service guideline TSB-110 “Biocides for Disinfection and Storage of Membrane Elements”.

9.7.3 SELECTION AND USE OF CLEANING CHEMICALS

There are a number of factors involved in the selection of a suitable cleaning chemical (or chemicals) and proper cleaning protocol. The first time you have to perform a cleaning, it is recommended to contact the manufacturer of the equipment, the RO element manufacturer, or a RO specialty chemical and service supplier. Once the suspected foulant(s) are identified, one or more cleaning chemicals will be recommended. These cleaning chemical(s) can be generic or can be private-labeled proprietary chemicals. Typically, the generic chemicals can be of technical grades and are available from local chemical supply companies. The proprietary RO cleaning chemicals can be more expensive, but may be easier to use and you cannot rule out the advantage of the intellectual knowledge supplied by these companies. Some independent RO service companies can determine the proper chemicals and cleaning protocol for your situation by testing at their facility a fouled element pulled from your system.

It is not unusual to use a number of different cleaning chemicals in a specific sequence to achieve the optimum cleaning. Typically, a high pH cleaning is used first to remove foulants like oil or biological matter, followed by a low pH cleaning to remove foulants like mineral scale or metal oxides/hydroxides fouling. There are times that order of high and low pH cleaning solutions is reversed or one solution only is required to clean the membranes. Some cleaning solutions have detergents added to aid in the removal of heavy biological and organic debris, while others have a chelating agent like EDTA added to aid in the removal of colloidal material, organic and biological material, and sulphate scale. An important thing to remember is that the improper selection of a cleaning chemical, or the sequence of chemical introduction, can make the foulant worse.

It is highly recommended that the membrane system operator thoroughly investigate the signs of fouling before they select a cleaning chemical and a cleaning protocol. Some forms of fouling (iron deposits and scaling commonly associated with well waters) may require only a simple low pH cleaning. However, for most complex fouling phenomena, following is recommended:

Flushing with permeate with addition of non-oxidizing biocide (DBNPA or similar type) at the end of the flushing.

9.7.3.1 High pH CIP
9.7.3.2 Flushing with permeate until pH on the brine side is below pH 8.5
9.7.3.3 Low pH CIP
9.7.3.4 Acid flushing with permeate and non-oxidizing biocide (DBNPA or similar type)

9.7.4 GENERAL PRECAUTIONS IN CLEANING CHEMICAL SELECTION AND USAGE

If you are using a proprietary chemical, make sure the chemical has been qualified for use with your membrane by the chemical supplier. The chemical supplier's instructions should not be in conflict with recommended cleaning parameters and limits listed in this Technical Service Guideline.

- If you are using generic chemicals, make sure the chemical has been qualified for use with your membrane in this Technical Service Guideline.
- Use the least harsh cleaning regiment to get the job done. This includes the cleaning parameters of pH, temperature, and contact time. This will optimize the useful life of the membrane.
- Clean at the recommended target temperatures to optimize cleaning efficiency and membrane life.
- Use the minimal amount of chemical contact time to optimize membrane life.
- Be prudent in the adjustment of pH at the low and high pH range to extend the useful life of the membrane. A "gentle" pH range is 4 to 10, while the harshest is 2 to 12.
- Oil and biologically -fouled membranes should not use a low pH clean-up first as the oil and biological matter will congeal.
- Cleaning and flushing flows should be in the same direction as the normal feed flow to avoid potential telescoping and element damage.
- When cleaning a multi-stage RO, the most effective cleaning is one stage at a time so cleaning flow velocities can be optimized and foulants from upstream stages don't have to pass through down-stream stages.
- Flushing out detergents with higher pH permeate can reduce foaming problems.
- Verify that proper disposal requirements for the cleaning solution are followed.
- If your system has been fouled biologically, you may want to consider the extra step of introducing a sanitizing biocide chemical before and after a successful cleaning. Biocides can be introduced before and immediately after cleaning, periodically (e.g. once a week), or continuously during service. You must be sure that the biocide is compatible with the membrane, does not create any health risks, is effective in controlling biological activity, and is not cost prohibitive.
- For safety reasons, make sure all hoses and piping can handle the temperatures, pressures and pH's encountered during a cleaning.
- For safety reasons, always add chemicals slowly to an agitated batch of make-up water.
- For safety reason, always wear safety glasses and protective gear when working with chemicals.

- For safety reasons, don't mix acids with caustics. Thoroughly rinse the 1st cleaning solution from the RO system before introducing the next solution

9.7.5 SELECTING A CLEANING SOLUTION

Below shown table lists the recommended generic chemical solutions for cleaning an RO membrane element based on the foulants to be removed.

IMPORTANT:
It is recommended that the MSDS of the cleaning chemicals be procured from the chemical supplier and that all safety precautions be utilized in the handling and storage of all chemicals.

RECOMMENDED CHEMICAL CLEANING SOLUTIONS

FOULANT	GENTLE CLEANING SOLUTION	HARSHER CLEANING SOLUTION
Calcium carbonate scale	1	4
Calcium, barium or strontium sulfate scale	2	4
Metal oxides/hydroxides (Fe, Mn, Zn, Cu, Al)	1	5
Inorganic colloidal foulants	1	4
Mixed Inorganic/organic colloidal foulants	2	6
Polymerized silica coating	None	7
Biological matter	2 or 3	6
NOM organic matter (naturally occurring)	2 or 3	6

RECIPES FOR CLEANING SOLUTIONS

"Recipes for Cleaning Solutions" offers instructions on the volumes of bulk chemical to be added to 379 litres of make-up water. Prepare the solutions by proportioning the amount of chemicals to the amount of make-up water to be used. Make-up water quality should be of RO permeate or deionized (DI) quality, and be free of chlorine and hardness. Before forwarding the cleaning solution to the membranes, it is important to thoroughly mix it, adjust the pH according to the target pH, and stabilize the temperature at the target temperature. Unless otherwise instructed, the cleaning design parameters are based on a chemical recirculation flow period of one hour and an optional chemical soak period of one hour.

Below table also shows “Maximum pH and Temperature Limits for Cleaning” highlights the maximum pH and temperature limits for specific membranes, after which irreparable membrane damage can occur. A suggested minimum temperature limit is 70 F (21 C), but cleaning effectiveness and the solubility of the cleaning chemical is significantly improved at higher temperatures

9.7.6 DESCRIPTION OF CLEANING SOLUTIONS

Note: The notation (w) denotes that the diluted chemical solution strength is based on the actual weight of the 100% pure chemical or active ingredient.

Solution 1: This is a low pH cleaning solution of 2.0% (w) citric acid (C6H8O7). It is useful in removing inorganic scale (e.g. calcium carbonate, calcium sulfate, barium sulfate, strontium sulfate) and metal oxides/hydroxides (e.g. iron, manganese, nickel, copper, zinc), and inorganic-based colloidal material. Note: Citric acid is available as a powder.

Solution 2: This is a high pH cleaning solution (target pH of 10.0) of 2.0% (w) of STPP (sodium tripolyphosphate) (Na5P3O10) and 0.8% (w) of Na-EDTA (sodium salt of ethylaminediaminetetraacetic acid). It is specifically recommended for removing calcium sulfate scale and light to moderate levels of organic foulants of natural origin. STPP functions as an inorganic-based chelating agent and detergent. Na-EDTA is an organic-based chelating cleaning agent that aids in the sequestering and removal of divalent and trivalent cations and metal ions. STPP and Na-EDTA are available as powders.

Solution 3: This is a high pH cleaning solution (target pH of 10.0) of 2.0% % (w) of STPP (sodium tripolyphosphate) (Na5P3O10) and 0.025% (w) Na-DDBS (C6H5(CH2)12-SO3Na) (sodium salt of dodecylbenzene sulfonate). It is specifically recommended for removing heavier levels of organic foulants of natural origin. STPP functions as an inorganic-based chelating agent and detergent. Na-DDBS functions as an anionic detergent.

Solution 4: This is a low pH cleaning solution (target pH of 2.5) of 0.5% (w) of HCL (hydrochloric) acid. It is useful in removing inorganic scale (e.g. calcium carbonate, calcium sulfate, barium sulfate, strontium sulfate and metal oxides/hydroxides (e.g. iron, manganese, nickel, copper, zinc) and inorganic-based colloidal material. This cleaning solution is considered to be harsher than Solution 1. HCL acid, a strong mineral acid, is also known as muriatic acid. HCL acid is available in a number of concentrations: (18 0 Baume = 27.9%), (20 0 Baume = 31.4%), (22 0 Baume = 36.0%).

Solution 5: This is a lower pH cleaning solution (natural pH is between pH 4 and 6. No pH adjustment is required) 1.0% (w) of $Na_2S_2O_4$ (sodium hydrosulfite). It is useful in the removal of metal oxides and hydroxides (especially iron fouling), and to a lesser extent calcium sulfate, barium sulfate and strontium sulfate. Sodium hydrosulfite is strong reducing agent and is also known as sodium dithionite. The solution will have a very strong odor so proper ventilation is required. Sodium hydrosulfite is available as a powder.

Solution 6: This is a high pH cleaning solution (target pH of 11.5) of 0.1% (w) of NaOH (sodium hydroxide) and 0.03% (w) of SDS (sodium dodecylsulfate). It is useful in the removal of organic foulants of natural origin, colloidal foulants of mixed organic/inorganic origin, and biological material (fungi, mold, slimes and biofilm). SDS is a detergent that is an anionic surfactant that will cause some foaming. This is considered to be a harsh cleaning regiment.

Note: Do not exceed maximum pH and temp limits for specific elements.

Solution 7: This is a high pH cleaning solution (target pH of 11.5) of 0.1% (w) of NaOH (sodium hydroxide). It is useful in the removal of polymerized silica. This is considered to be a harsh cleaning regiment.

Note: Do not exceed maximum pH and temp limits for specific elements.

Important:

It is recommended that the MSDS of the cleaning chemicals be procured from the chemical supplier and that all safety precautions be utilized in the handling and storage of all chemicals.

RECIPES FOR CLEANING SOLUTIONS

The quantities listed below are to be added to 300 litres of dilution water.

BULK INGREDIENTS	QUANTITY	TARGET 1 pH ADJUSTMENT	TARGET 1 TEMP.
Citric acid (as 100% powder)	7.7 kg	No pH adjustment is Required	(40˚C)
STPP (sodium tripolyphosphate) (as 100% powder) **Na-EDTA** (Versene 220 or equal) (as 100% powder)	7.7 kg 3.18 kg	Adjust to pH 10.0 with sulfuric or hydrochloric acid.	(40˚C)
STPP (sodium tripolyphosphate) (as 100% powder) **Na-DDBS** Na-dodecylbenzene sulfonate	7.7 kg 0.1 kg	Adjust down to pH 10.0 with sulfuric or hydrochloric acid.	(40˚C)
HCl acid (hydrochloric acid (as 220 Baume or 36% HCL)	1.78 litres	Slowly adjust pH down to 2.5 with HCL acid. Adjust pH up with sodium hydroxide.	(35˚C)
Sodium hydrosulfite (as 100% powder)	3.86 kg	No pH adjustment is required.	(35˚C)
NaOH (sodium hydroxide) (as 100% powder) (or as 50% liquid)	0.38 kg 0.49 liters	Slowly adjust pH up to 11.5 with sodium hydroxide. Adjust pH down to 11.5 by adding HCL acid	(30˚C)
NaOH (sodium hydroxide**)** (as 100% powder) (or as 50% liquid) **SDS** (sodium dodecylsulfate)	0.38 kg 0.49 liters 0.11 kg	Slowly adjust pH up to 11.5 with sodium hydroxide. Adjust pH down to 11.5 by adding HCL acid.	(30˚C)

Note: Cleaning operations performed at the extremes may result in a more effective cleaning, but can shorten the useful life of the membrane due to hydrolysis. To optimize the useful life of a membrane, it is recommended to use the least harsh cleaning solutions and minimize the contact time whenever possible. The pH of the feed stream or cleaning solution should be closely monitored and controlled. The pH meters used to measure and control pH should be regularly calibrated to ensure accuracy. It is typical to re-circulate cleaning chemicals through the RO for 1 hour. At the pH limits shown above, cleaning exposure at temperatures less than 40 C is limited to 60 minutes, at temperatures greater than 40 C exposure is limited to 30 minutes. Extended soaking is possible, but at less aggressive pH levels.

9.7.7 RO MEMBRANE FOULING AND CLEANING

RO Membrane Element Cleaning and Flushing Procedures

The RO membrane elements can be cleaned in place in the pressure tubes by recalculating the cleaning solution across the high-pressure side of the membrane at low pressure and relatively high flow. A cleaning unit is needed to do this. RO cleaning procedures may vary dependent on the situation. The time required to clean a stage can take from 4 to 8 hours.

A general procedure for cleaning the RO membrane elements is as follows:

9.7.7.1 Perform a low pressure flush at 60 psi (4 bar) or less of the pressure tubes by pumping clean water from the cleaning tank (or equivalent source) through the pressure tubes to drain for several minutes. Flush water should be clean water of RO permeate or DI quality and be free of hardness, transition metals, and chlorine.

9.7.7.2 Mix a fresh batch of the selected cleaning solution in the cleaning tank. The dilution water should be clean water of RO permeate or DI quality and be free of hardness, transition metals, and chlorine. The temperature and pH should be adjusted to their target levels.

9.7.7.3 Circulate the cleaning solution through the pressure tubes for approximately one hour or the desired period of time. At the start, send the displaced water to drain so you don't dilute the cleaning chemical and then divert up to 20% of the most highly fouled cleaning solution to drain before returning the cleaning solution back to the RO Cleaning Tank. For the first 5 minutes, slowly throttle the flow rate to 1/3 of the maximum design flow rate. This is to minimize the potential plugging of the feed path with a large amount of dislodged foulant.. For the second 5 minutes, increase the flow rate to 2/3 of the maximum design flow rate, and then increase the flow rate to the maximum design flow rate. If required, readjust the pH back to the target when it changes more than 0.5 pH units.

9.7.7.4 An optional soak and recirculation sequence can be used, if required. The soak time can be from 1 to 8 hours depending on the manufacturer's recommendations.

Caution should be used to maintain the proper temperature and pH. *Do not exceed maximum pH and temperature limits for specific elements.* Also note that this does increase the chemical exposure time of the membrane.

9.7.7.5 Upon completion of the chemical cleaning steps, a low pressure Cleaning Rinse with clean water (RO permeate or DI quality and free of hardness, transition metals, and chlorine) is required to remove all traces of chemical from the Cleaning Skid and the RO Skid. Drain and flush the cleaning tank; then completely refill the Cleaning Tank with clean water for the Cleaning Rinse. Rinse the pressure tubes by pumping all of the rinse water from the Cleaning Tank through the pressure tubes to drain. A second cleaning can be started at this point, if required.

9.7.7.6 Once the RO system is fully rinsed of cleaning chemical with clean water from the Cleaning Tank, a Final Low Pressure Clean-up Flush can be performed using pretreated feed water. The permeate line should remain open to drain. Feed pressure should be less than 60 psi (4 bar). This final flush continues until the flush water flows clean and is free of any foam or residues of cleaning agents. This usually takes 15 to 60 minutes. The operator can sample the flush water going to the drain for detergent removal and lack of foaming by using a clear flask and shaking it. A conductivity meter can be used to test for removal of cleaning chemicals, such that the flush water to drain is within 10-20% of the feed water conductivity. A pH meter can also be used to compare the flush water to drain to the feed pH.

9.7.7.7 Once all the stages of a train are cleaned, and the chemicals flushed out, the RO can be restarted and placed into a Service Rinse. The RO permeate should be diverted to drain until it meets the quality requirements of the process (e.g. conductivity, pH, etc.). It is not unusual for it to take from a few hours to a few days for the RO permeate quality to stabilize, especially after high pH cleanings.

10.0 REPLACEMENT OF FILTER CARTRIDGE (CF – 201)

10.1 CRITERIA FOR REPLACEMENT OF CARTRIDGE FILTER:

10.1.1 If pressure difference (ΔP) across cartridge filter increases above **0.7 Kg/cm^2**

10.2 REPLACEMENT FREQUENCY:

10.2.1 1 Month ± 1 week or above mentioned criteria, whichever is earlier

10.3 PROCEDURE:

10.3.1 **STOP** ‘PW GENERATION SYSTEM’ if system is operational in ‘**AUTO’** Mode or STOP all in-line pumps, dosing pumps if in ‘**MANUAL’** Mode

10.3.2 After Ensuring that the pressure in the cartridge filter is zero (check in PG – 201 & 202), dismantle piping from outlet nozzle

10.3.3 Open the cartridge filter cover through TC clamp which is holding body cover with base.

10.3.4 Remove the existing filter elements and clean them

10.3.5 Drain the cartridge filter body and wash it with water

10.3.6 Install new filter cartridge elements and prefix the covers tightening TC clamp

10.3.7 Connect upstream piping to outlet nozzle

10.3.8 Meanwhile check pressure across the cartridge filter (CF – 201) by checking inlet (PG – 201) and outlet pressure (PG – 202). If differential pressure is < 0.7 Kg/cm^2, then consider that installed cartridges are working properly

10.3.9 Check for any visual leakages. If not that means piping is connected properly

10.3.10 Now system is ready to operate in ‘AUTO’ mode or ‘MANUAL’ mode as desired.

Record replacement of cartridge filters in equipment log sheet

11.0 CLEANING OF FILTER CARTRIDGE (CF – 201)

NOTE

If cartridge for replacement is not available in market then cleaning procedure to be perform.

11.1 CLEANING FREQUENCY:

11.1.1 After 1 month ± 7 days

11.2 PROCEDURE:

11.2.1 STOP 'PW GENERATION SYSTEM' if system is operational in 'AUTO' Mode or STOP all in-line pumps, dosing pumps if in 'MANUAL' Mode

11.2.2 After Ensuring that the pressure in the cartridge filter is zero (check in PG – 201 & 202), dismantle piping from outlet nozzle

11.2.3 Open the cartridge filter cover through TC clamp which is holding body cover with base.

11.2.4 Remove the existing filter elements

11.2.5 Drain the cartridge filter body and wash it with water

11.2.6 Install cleaned cartridge elements back into housing and prefix the covers tightening TC clamp

11.2.7 Connect upstream piping to outlet nozzle

11.2.8 Meanwhile check pressure across the cartridge filter (CF – 201) by checking inlet (PG – 201) and outlet pressure (PG – 202). If differential pressure is as less than 0.7 Kg/cm^2, then consider that installed cartridges are working properly

11.2.9 Check for any visual leakages. If not that means piping is connected properly

11.2.10 Now system is ready to operate in 'AUTO' mode or 'MANUAL' mode as desired.

NOTE

Record cleaning of cartridge filters in equipment log sheet

12.0 CIP OF EDI SYSTEM

12.1 CRITERIA FOR CIP

12.1.1 If the product differential pressure increases by 50% without a change in temperature and flow, or

12.1.2 The reject differential pressure increases by 50% without a change in temperature and flow,

12.1.3 The product quality declines without a change in temperature, flow, or feed conductivity, or

12.1.4 The module's electrical resistance increases by 25% without a change in temperature.

12.2 CIP FREQUENCY:

12.2.1 **6 months ± 1 week** OR above criteria specified in point no. 12.1 whichever is earlier

12.3 SAFETY PRECAUTIONS

Safety precautions shall be followed while doing chemical cleaning of EDI membranes. In addition to maintaining adequate ventilation, personal protective equipment shall be utilized to prevent injury or personnel hazards.

1.1 CLEANING REQUIREMENT & PROCEDURE

REFER THE EDI MANUAL FOR CHEMICALS REQUIRED & DETAILED PROCEDURE FOR THE CIP OF EDI SYSTEM

13.0 CLEANING OF RO FEED TANK

13.1 CLEANING FREQUENCY:

6 months ± 1 week

13.2 PROCEDURE:

13.2.1 **Close** outlet valve of the tanks

13.2.2 Check the water level of the tank

13.2.3 Add **sodium hypochlorite (6% w/v concentration - approx.)** in the tank manually & stir with the help of the stainless steel ladder to make the solution of approx. 50 ppm

13.2.4 Hold on this condition for minimum 5-6 hrs or overnight if possible

13.2.5 Check the concentration of the solution. Top up with chemical if concentration drops down below 50 ppm during hold time after every 2 hour

13.2.6 After completion of holding period drain out the water from tank completely using drain valve

13.2.7 Fill the tank with fresh water up to $1/3^{rd}$ level. Stir this water with Stainless steel rod wounded with the clean soft cloth at the rod tip. Ensure that it cleans the bottom of tank properly

13.2.8 Rub the bottom surface of the tank using this rod & drain out the water from the tank completely

13.2.9 Repeat the process specified above till the bottom surface of the tank get cleaned from the chemical deposition

14.0 MAINTENANCE

Refer **ANNEXURE – 2** 'Maintenance Schedule' for preventive maintenance of major equipments and critical instruments.

Every Breakdown of Equipment should be recorded in Equipment History Card.